

Traffic Flow Prediction Using LSTM, ARIMA & SARIMA Model for Intelligent Transportation System

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Abstract— Accurate and timely prediction of traffic data is an essential component in the planning, administration, and control of urban traffic in Intelligent Transportation Systems (ITS). Our study is dedicated to exploring three distinct models: LSTM (Long Short Term Memory), ARIMA (Autoregressive Integrated Moving Average) with parameters (p, d, q) that specify autoregression, differencing, and moving average orders, and SARIMA (Seasonal Autoregressive Integrated Moving Average), intended for analyzing seasonal time series data. ARIMA and SARIMA models are conventional statistical techniques employed for the study of time series data. They provide reliable methodologies for forecasting future traffic patterns through the analysis of historical data. To choose the most efficient model among the three options, we utilize performance measurement procedures to evaluate their predictive abilities. We utilize an external dataset obtained from the U.S. Government Open Repository. This dataset is classified as geographical data. Upon analysis, it was shown that all three SARIMA models had superior performance.

Keywords—Intelligent Transportation System, ARIMA, SARIMA, LSTM

I. INTRODUCTION

The Internet of Things (IoT) is a groundbreaking technological advancement that links our tangible surroundings with the virtual realm. The system consistently produces and analyses many types of data, which might be unstructured or semi- structured, such as audio and video streams. Yet, the widespread distribution of data via Intelligent Transportation Systems (ITS) presents notable difficulties in data structuring, network accessibility, and bandwidth administration. The intricate equilibrium between network resources and the requirements of time-sensitive data becomes pivotal, since any deficiency might result in delays and malfunctions. Traffic flow forecasting is becoming increasingly important in the field of intelligent transportation. The goal is to resolve present roadblocks in transportation that are identified through dynamic feedback and automation. ITS aims to tackle complex problems like accessibility, congestion, pollution, safety,

security, noise, and resource consumption by combining various information technologies such as control systems, computing, communication, algorithms, databases, models, and human interfaces. Although the transportation industry is mostly responsible for contributing to global warming through the release of greenhouse gases, it is indisputable that mobility is crucial in both societal and economic contexts. The rise in the number of vehicles unquestionably improves everyday living, but at the same time worsens traffic congestion, necessitating the development of reliable traffic flow forecast technology. Utilizing extensive traffic data obtained through sensor networks, traffic flow prediction plays a crucial role in reducing congestion and avoiding economic losses caused by long commute times. The primary objective of Intelligent Transportation Systems is to enhance traffic control and population surveillance, consequently strengthening risk evaluation skills. Yet, the intricate and diverse nature of transportation networks provide significant obstacles in fulfilling real-time demands. The emergence of advanced technologies like Big Data, IoT, and Artificial Intelligence offers a chance to transform conventional industrialized cities by incorporating different urban components, such as healthcare, hospitality, transportation, power, and public safety. Inadequate and perilous traffic management in recent years has led to a significant rise in both injuries and property damage. The notion of smart cities, which transform urban environments into digital societies, highlights the crucial significance of Intelligent Transportation Systems (ITS) in improving human mobility, safety, and comfort. ITS provides customers with up- to-date traffic information, nearby amenities, train timetables, and seat availability, resulting in shorter journey times and improved transportation efficiency. Within the domain of Intelligent Transportation Systems (ITS) development, several methodologies contribute to advancement. Out of these options, VDTN is particularly well- suited for several applications like intelligent transportation systems, smart cities, marketing and advertising, telemedicine, multimedia data delivery, and internet consumption [8]. Vehicular Delay Tolerant Networks (VDTNs) are utilized for communications that do not rely on end-to-end connection[9]. Instances of frequent disconnections and significant delays are prevalent in Vehicular ad-hoc networks.[10]. VDTNs use the store carry and forward paradigm from DTN to forward the messages to the destination[11]. The VDTN system may be classified into two types of nodes: fixed nodes and mobile nodes.[12].

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The Ministry of Electronics and Information Technology (MEITY) in India, founded on July 19, 2016, leads the domestic creation of Smart Transportation Systems (ITS) solutions through the INTRANSE-11 initiative. This research and development project, supported by MEITY, involved the collaboration of major players such as Mahindra & Mahindra, IIT-Madras, IIT-Bombay, and IISE-Bangalore at a national level. The program seeks to tackle India's distinct traffic difficulties by implementing projects such as On-Board Driver Assistance and Warning Systems (ODAWS), bus signal priority systems, and the Common Smart IoT Connective (COSMIC) software. The INTRANSE-11 initiative represents a notable advancement in utilizing state-of-the-art technology to transform transportation networks and facilitate the development of intelligent and environmentally-friendly communities.

II. RELATED WORK

Significant issues arise in the identification, categorization, and prediction of future traffic flow in transportation systems, which has prompted much study into different prediction methodologies. Below, we provide a compilation of approaches that have been investigated in recent research: He-Xuan Hu and his team developed the Multi-layer Transformer and Deep Learning for Traffic Flow Prediction model, which integrates transformer and deep learning techniques. This method utilizes many encoders and decoders to autonomously extract characteristics from the initial traffic data. The output obtained from the experiments conducted on both weekday and holiday datasets provide highly favorable performance metrics. RMSE for the dataset including weekdays is 0.191, MAE is 0.165. vacation dataset reports an RMSE of 0.227 and an MAE of 0.192. To provide the utmost precision in forecasting, a linear model is utilized in order to combine prediction scores from both datasets [1]. Peng et al. concentrate on the real-time monitoring of population utilizing streams of transportation big data. The authors put up suggestions for scheduling flows in time-triggered and audio-video bridging systems. An online scheduling approach called RFSD is introduced for TT flows, utilizing lion swarm optimization to assign priorities. The experimental findings show a significant improvement of 22% in effective scheduling when compared to the PAS Algorithm. Furthermore, there is a significant decline of 24% in the average delay and a 27% drop in the variability of data transmission [2]. Ke Zhao and colleagues introduce the Improved Dung Beetle Optimization -LSTM model, which integrates LSTM with variational mode decomposition to forecast traffic flow. The VMD-IDBO-LSTM model exhibits a decrease in MAE of 26.69 and 7.5108, a decrease in MAPE of 8.08059% and 2.27569%, and a decrease in RMSE of 33.6912 and 8.7657, respectively, when compared to the LSTM and VMD-LSTM models. These reductions indicate an enhanced level of accuracy in predicting short-term traffic flow [3]. Vienna N. Katambire et al. focus on utilizing LSTM and ARIMA models to accurately predict real-time traffic Counts at Mahima Junction Kagali. The performance measures, namely MAE, MAPE, and RMSE, demonstrate The LSTM

model is the most effective choice for predicting monthly traffic flow. [4]. In this study, Yuyan Annie Peng and her colleagues propose the FD-Markov-LSTM model, which integrates the basic diagram, Markov chain, and LSTM models to provide precise traffic flow prediction. The FD-Markov LSTM model has higher performance in comparison to benchmark models like as Markov, ARIMA, KNN, Random Forest, and LSTM models. The comparison research reveals that the FD-Markov LSTM model yields substantial improvements, with a decrease of around 39% in MAE, 35% in RMSE, and 74% in MAPE [5].

III. METHODOLOGY AND DATA SOURCE

We have selected three models to forecast the movement of vehicles in transportation networks. Firstly, we commence the training process on our dataset, followed by the evaluation of the LSTM, ARIMA, and SARIMA models. This evaluation involves projecting the traffic flow for upcoming days. Ultimately, we assess the findings of the models in order to provide precise and effective outputs.

A. Data Collection

The traffic flow counts dataset was obtained from the U.S. Government open dataset repository, data.gov. The dataset is not provided by the federal government and includes information on the volume of traffic from 2018. The metadata for this dataset was generated on September 2, 2022, and subsequently revised on September 15, 2023. The supplementary information includes the publisher, which is City of Seattle ArcGIS Online, the dataset maintainer, which is Transportation Seattle city GIS, the metadata type, which is classed as GeoSpatial, and the access level, which is set to public.

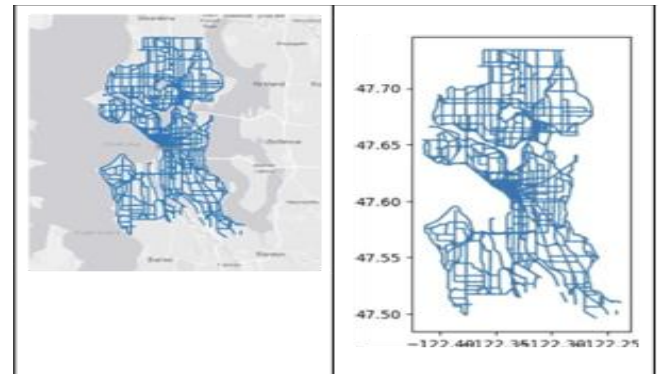


Figure 1. Geo-spatial Area of Seattle

B. Data Preprocessing

We have examined traffic flow data from a certain geographic area and transformed it into a suitable format to extract more practical and useful information. During preprocessing, we modify the characteristics and target variables to align with our unique requirements. The characteristics we have are 'DOWNTOWN' and 'AWDT_ROUNDED', and our goal is to forecast the target variable 'AWDT'.

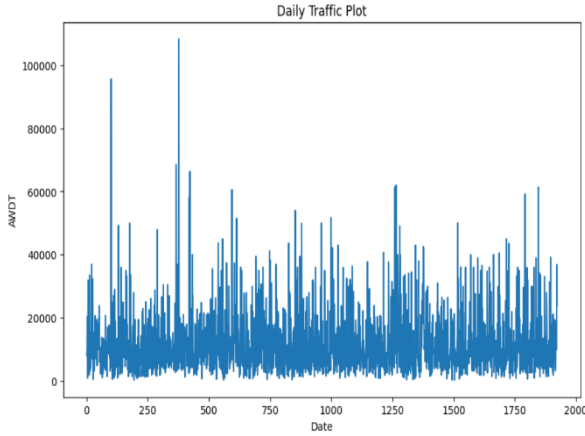


Figure.2.Daily Traffic Plot based on Average Week Day Daily Traffic

C. Feature Selection

Feature selection is a crucial component of machine learning as it allows for the identification of the most pertinent characteristic from our dataset in order to improve model performance and reduce overfitting. Within our dataset, there are 14 features- Downtown, StartDate, DataQuality, Flowsegid, Objectid, Compkey, Pmpk, Ampk, Awdt, Awdtrounded, Sname_ord, Flags and Shape_Length.

Table.1. Attribute name and Their Meaning

Attribute_Name	Meaning
COMPKEY	Unique identifier for each row
STNAME_ORD	Street name or location identifier
FLOWSEGID	Flow segment ID
DOWNTOWN	Indicates if the location is in the downtown area(Y/N)
START_DATE	Date and Time when the study started
AMPK_PMPK	Average and Peak traffic count
AWDT,ADT,AWDT_ROUNDED	Average week daily traffic
DATAQUALITY	Quality of the data
FLAGS	Flags or Additional information
SHAPE_LENGTH	Length of the shape(geospatial feature)

However, our analysis specifically concentrates on 10 characteristics. Firstly, we divide the dataset into two parts: the features (X) and the target variable (y). Afterwards, we utilize SelectKBest to identify the ideal features by using an appropriate scoring function, such as chi2 for classification or f_regression for regression. Next, we get feature scores and p-values from the fitted SelectKBest object and arrange them into a DataFrame to contain the names of the features, their scores, and their p-values. In order to determine the feature with the greatest impact, we organize the characteristics in a decreasing order according to their ratings, making it easier to identify the most important feature.

D. Proposed Model Architecture

Within our suggested architecture “Fig. 3”, the model's workflow starts by supplying inputs, which encompass our characteristics. The LSTM model analyzes these inputs in its first phase, resulting in the generation of residual plots.

These residual plots are then employed as inputs for both the ARIMA and SARIMA models. Upon completion of the data processing, which includes both training and testing stages, these models generate predicted values. Afterwards, these predicted values are combined with the output from the LSTM model, resulting in the ultimate output.

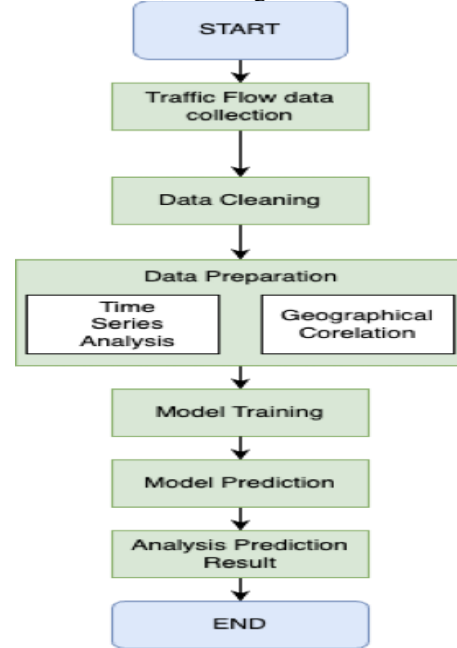


Figure.3.Proposed Model Architecture

E. Applied Models

These models initially examine historical data, use past data to generate predictions. In our research, we implement the Long Short Term Memory, AutoRegressive Integrated Moving Average and Seasonal AutoRegressive Integrated Moving Average models. SARIMA is an enhanced model that extends the capabilities of the ARIMA model.

(a) Long Short Term Memory Model:

A Recurrent Neural Network (RNN) may acquire knowledge of the inherent temporal connections in past data by iteratively examining time-series data. Algorithmic defects that arise during the back-propagation process can result in the vanishing or exploding gradient problem, which impairs the capacity to learn sequences from distant time points. The issue persisted without resolution until the implementation of the LSTM model. The user's text is [6]. The LSTM model operates on five key principles: long-term memory, short-term memory, forget gate, input gate, and output gate. Short-term memory effectively preserves relevant information, but the forget gate eliminates superfluous data. The input gate retains relevant information, whereas the output gate produces expected outcomes.

(b) Auto Regressive Integrated Moving Average Model:

ARIMA was proposed as a solution for traffic prediction to overcome the constraint of the ARIMA model in dealing with non-linear traffic data. The text is enclosed in tags.

ARIMA is a time series forecasting model that incorporates AutoRegressive (AR), Integrated (I), and Moving Average (MA) elements. AR: It denotes the autoregressive component, which predicts future values by using linear combinations of past values. The sign "I" denotes the mathematical operation of integration, which is employed to convert a time series into a stationary process characterized by a consistent mean and variance. The moving average (MA) is the part of the forecasting equation that specifically addresses the residual components. The ARIMA model is characterized by the parameters p , d , and q .

Table.2.ARIMA Parameters

Parameters	Definition
p	Autoregressive order
d	Differencing order
q	Moving average order

ARIMA is well-suited for analyzing time series data that lacks seasonal patterns, on the assumption that past patterns will persist in the future. The data is derived from a time series that is not influenced by seasonal patterns, and it is presumed that the historical patterns will continue into the future.

(c) Seasonal AutoRegressive Integrated Moving Average Model:

SARIMA is a time series analysis model that builds upon the ARIMA model by specifically accounting for seasonality in the data. This model includes supplementary seasonal elements indicated by P , D , and Q , which respectively represent seasonal autoregressive, seasonal differencing, and seasonal average movement terms. The SARIMA notation is used to express a time series model as $SARIMA(P, D, Q)_s$. In this notation, (P, D, Q) represents the parameters for the seasonal component, and s determines the duration of the seasonal cycle. SARIMA is highly suitable for analyzing time series data that exhibit distinct and recognizable seasonal patterns.

IV. RESULT ANALYSIS

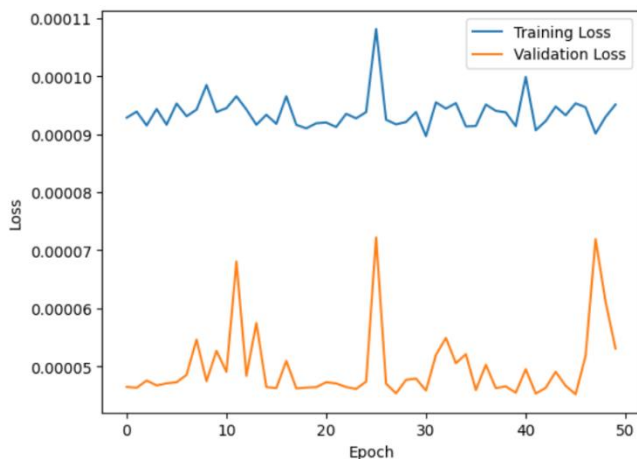


Figure.4. visualizing the training and validation loss

The figure presented in "Fig.4", demonstrates the variations in training and validation loss of a machine learning model during the training epochs. This visualization allows for the assessment of the model's performance and the identification of any problems, such as overfitting or underfitting. The machine learning model employs the training data, including X_{train} and y_{train} . The user specifies the training parameters, which consist of 50 epochs, a batch size of 32, and the validation data represented by X_{test} and y_{test} . The historical record of training is stored in the history object, which includes information on training loss, validation loss, and numerous metrics over the epochs. The figure's x-axis is denoted as 'Epoch', signifying the quantity of training. epochs. The vertical axis of the graph is designated as 'Loss', indicating the magnitudes of loss.

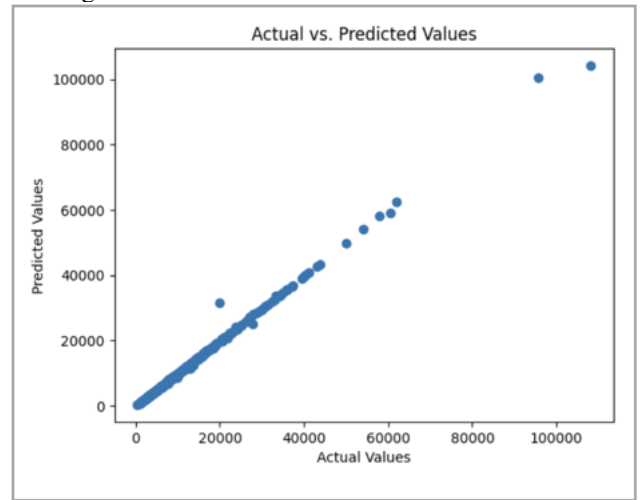


Figure.5. visualizing the performance of a regression model plotting actual target values resinsting the predicted target values.

"Fig.5", represents a scatter plot illustrating the comparison between the actual target values (y_{test_actual}) and the predicted target values (y_{pred}). Each point on the plot corresponds to a distinct data instance, where the x-coordinate represents the actual value and the y-coordinate represents the predicted value. This chart enables the assessment of the level of agreement between the model's forecasts and the observed data.

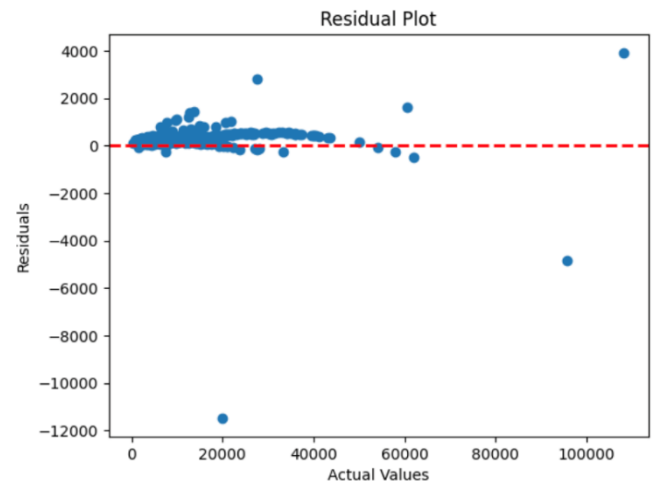


Figure.6. Analyze the residuals of a regression model

The “Fig.6”, is meant to analyze the residuals of a regression model. Residuals refer to the disparities between the observed (actual) values and the projected values. By plotting residuals against the actual values, we can examine whether the model makes consistent errors across the range of observations.

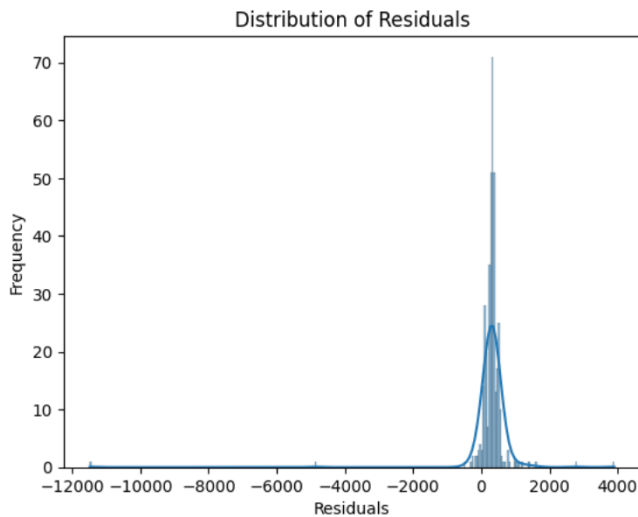


Figure.7. A histogram plot of the residuals

In histogram plot of the residuals is shown in “Fig 7”. The residuals variable encompasses the discrepancies between the observed and forecasted values. The inclusion of the kde=True option incorporates a kernel density estimation plot into the histogram. This graphic displays an estimation of the probability density function of the residuals. The x-axis label of the image is denoted as 'Residuals', signifying the residuals generated by the regression model. The y-axis label of the graphic is 'Frequency', denoting the frequency of occurrence for each residual value.

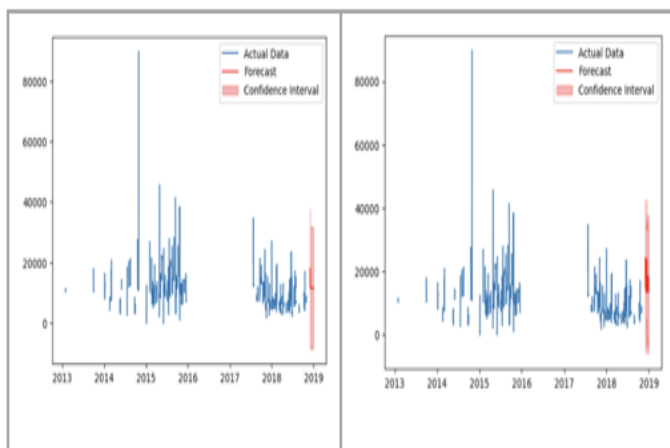


Figure.8. Time Series Analysis and Prediction Applying ARIMA and SARIMA Model.

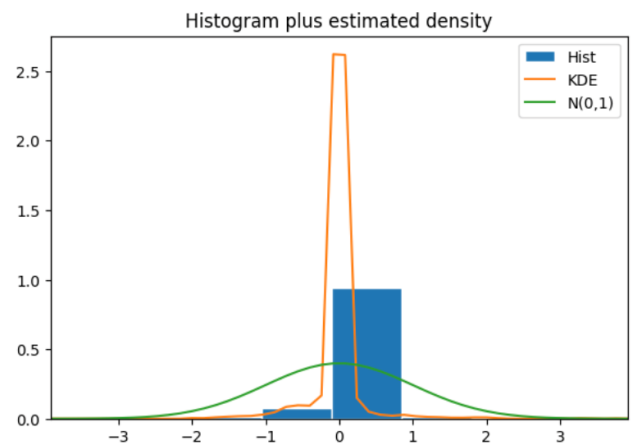


Figure.9. Histogram Plus Estimated Density

The Analysis includes a histogram of residuals , a Q-Q plot to access the normal distribution of residuals, a correlogram to examine autocorrelation and partial Autocorrelation of residuals, and a plot comparing the Time Series with the fitted values.

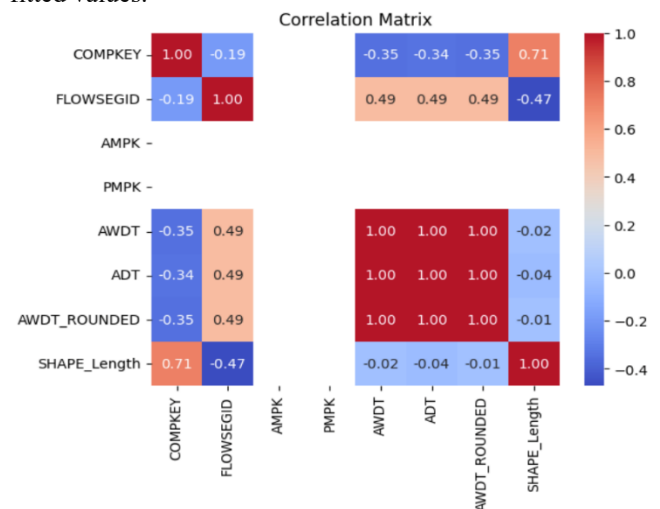


Figure.10. Correlation Matrix

The “Figure.10”, provides visualizations to explore relationships between variables in the dataset, including pairwise scatterplots and a heatmap of the correlation matrix. These visualizations help understand the data structure, identify patterns, and discover potential correlations between variables. A pair plot is a grid of scatterplots showing the relationships between pairs of variables in the DataFrame. This helps visualize potential correlations and distributions between different variables. Pair Plot A pair plot is a grid of scatterplots where each numerical variable in the dataset is paired with every other numerical variable. For example, if our dataset has four numerical variables, the pair plot will consist of a 4x4 grid of scatterplots. Each scatterplot inside the pair plot illustrates the correlation between two numerical variables. Each scatterplot's x-axis represents one variable, while the y-axis represents another one. Pair plots are valuable for visually examining connections between numerical data, discerning patterns, and finding possible outliers or clusters. Heatmap of the correlation matrix A correlation matrix is a symmetrical matrix that presents the associations

between pairs of numerical variables in a dataset. Each cell within the matrix represents the correlation coefficient between two variables. Correlation coefficients measure the size and direction of the linear relationship between two variables. The numerical range extends from -1 to 1, where a value of 1 indicates a perfect positive correlation, meaning that when one variable increases, the other variable increases in a comparable manner. A score of -1 indicates a strong negative correlation, indicating that as one variable increases, the other variable decreases in a perfectly consistent manner. A score of 0 indicates the lack of any linear correlation between the variables.

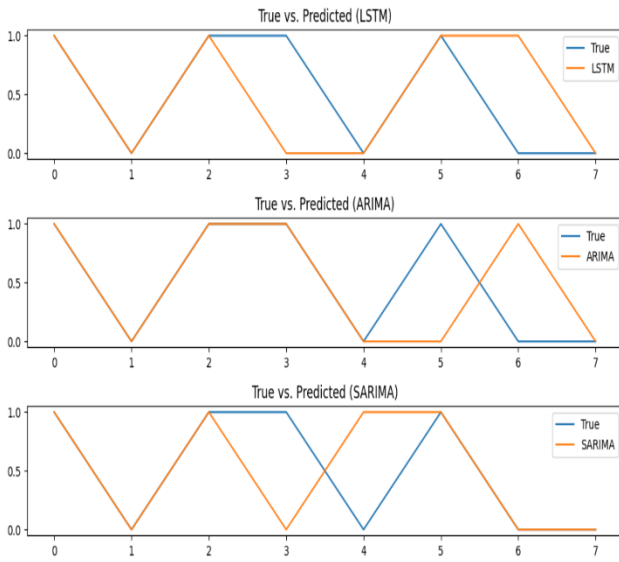


Figure.11. Model Prediction

The graph labeled as 'True'. Next, we graph the Forecasted values from LSTM model (predicted_labels_lstm) and label it as 'LSTM'. Enables the visualization of a legend on the plot, which provides information about the actual and expected values. The primary objective is to conduct a visual comparison of the predictive performance of three distinct models (LSTM, ARIMA, and SARIMA) versus the actual data. By creating distinct subplots and showing the real values against the expected values for each model. This comparison enables a qualitative assessment of the models' prediction capacities. One can make observations on the degree of similarity between the anticipated values and the real values, as well as identify any significant disparities or trends in the models' predictions. This comparison facilitates the selection of the best appropriate model for the particular forecasting assignment by evaluating its performance in relation to the others. The resulting graphic enables you to juxtapose the actual values with the anticipated values, therefore evaluating the performance of each model.

IV. CONCLUSION

Forecasting. This study entailed a juxtaposition between the model and LSTM, a deep learning methodology. The two utilized models have exhibited robust performance in forecasting future demand or trends over different time spans. Both models rely solely on previous data to comprehend the pattern and forecast the likely future. The

ARIMA model efficiently filters out high-frequency noise in the data and precisely forecasts development trends. There is a need for daily projections of time series data since we have detected a tendency of increasing traffic flow in the future. An assessment must be conducted to examine the diverse effects of heterogeneity and other previously unexplored factors that influence traffic. Understanding the importance of quantifying the uncertainty associated with deterministic predictions of traffic flow. Subsequent investigations should focus on developing a probabilistic forecasting model utilizing the proposed technique to effectively tackle the issue of congestion. After applying performance measurements to our models, we have determined that SARIMA outperforms other models in our dataset. Additionally, we have observed that our model yields varying outcomes when applied to other datasets. In the future, we may utilize the same algorithm to forecast weekly traffic patterns in order to alleviate traffic congestion.

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